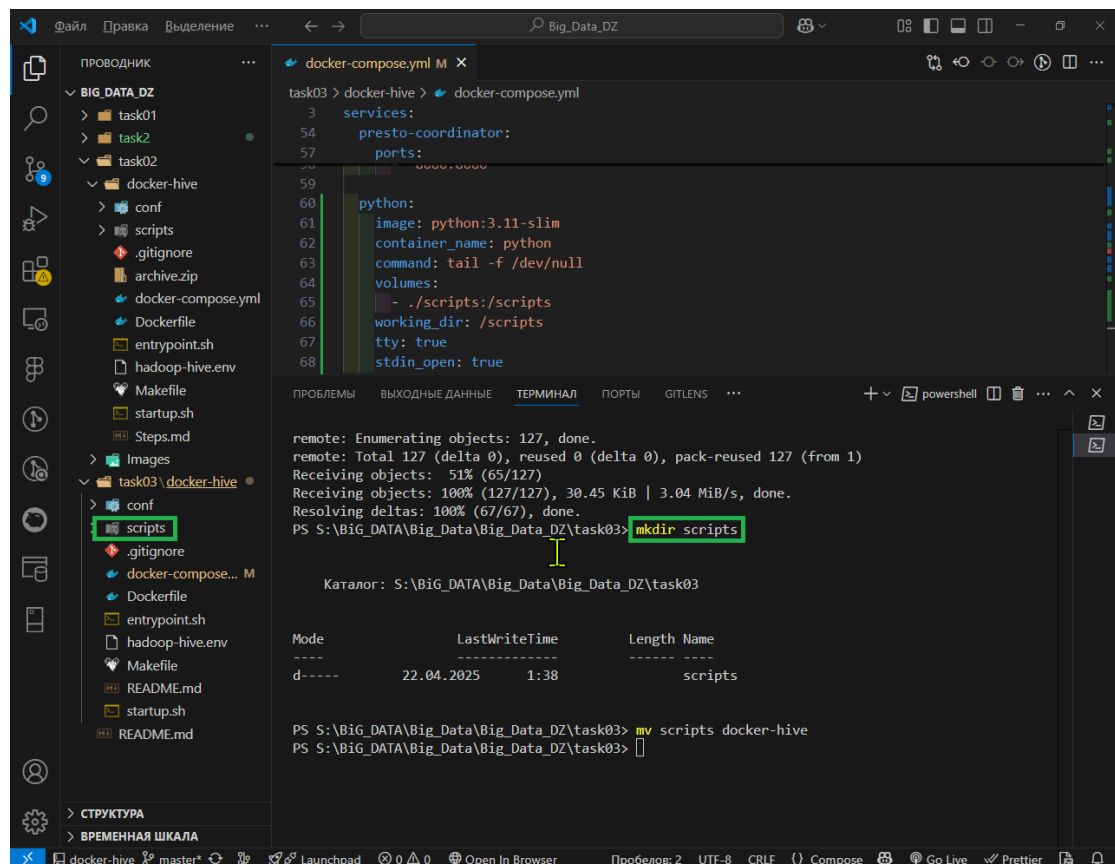
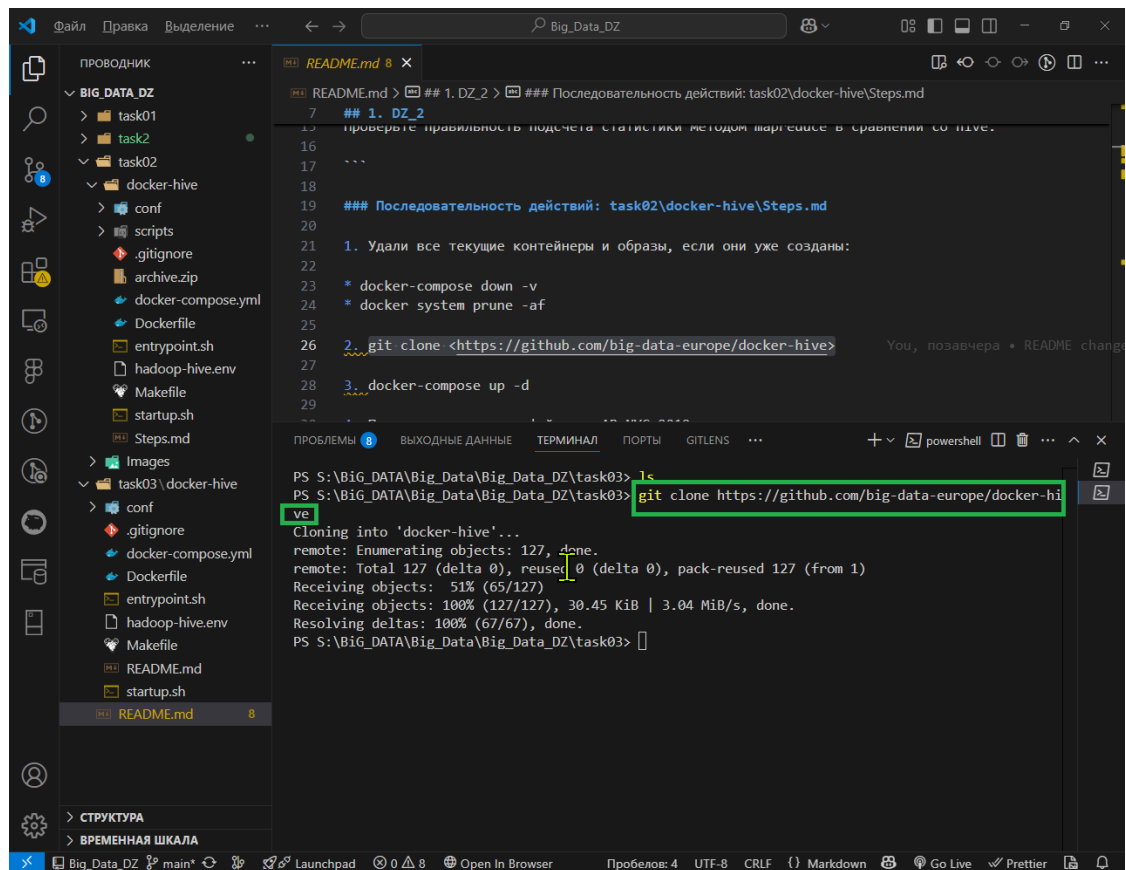
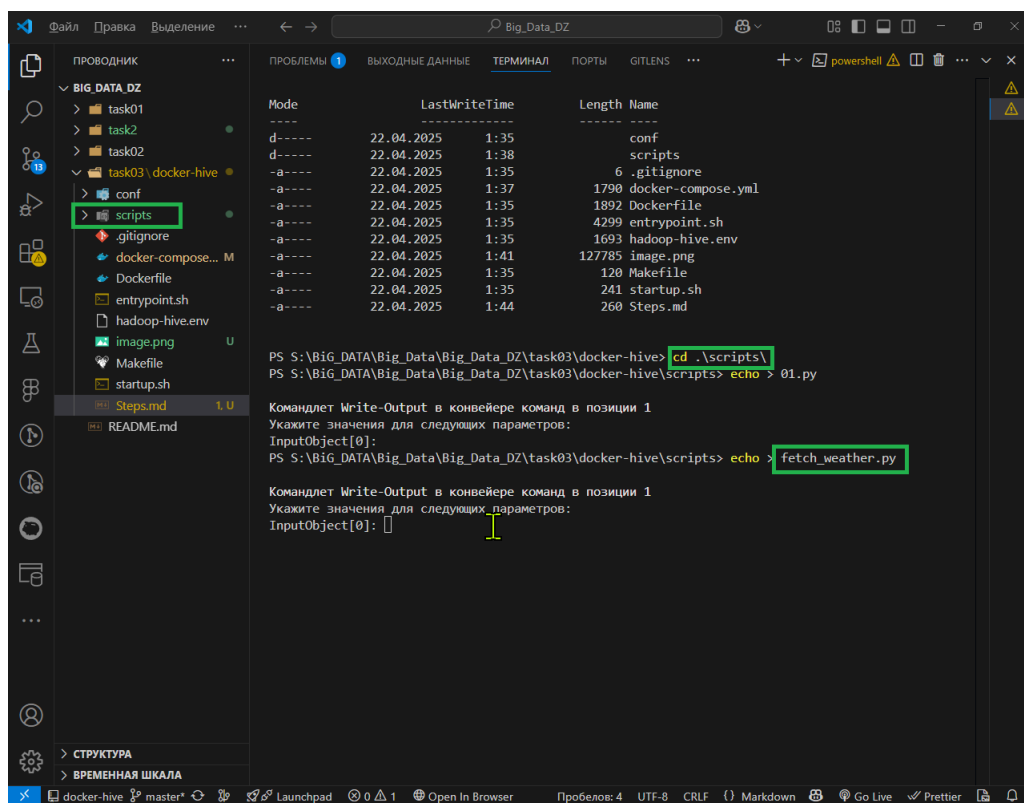
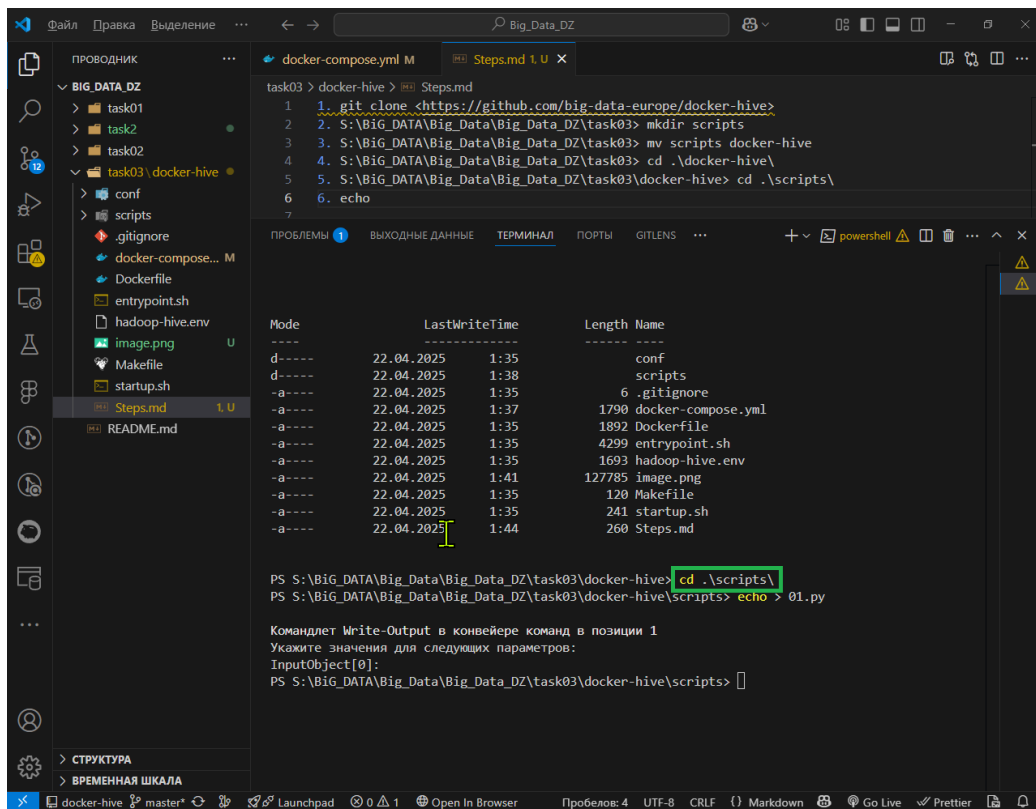






The screenshot shows the VS Code editor interface with the file explorer on the left displaying the project structure. The main editor window shows the `fetch_weather.py` script. The script imports `requests` and `pandas`, defines a list of cities with their coordinates, and fetches weather data from the Open-Meteo API. It then saves the data to a CSV file named `weather_data.csv`.

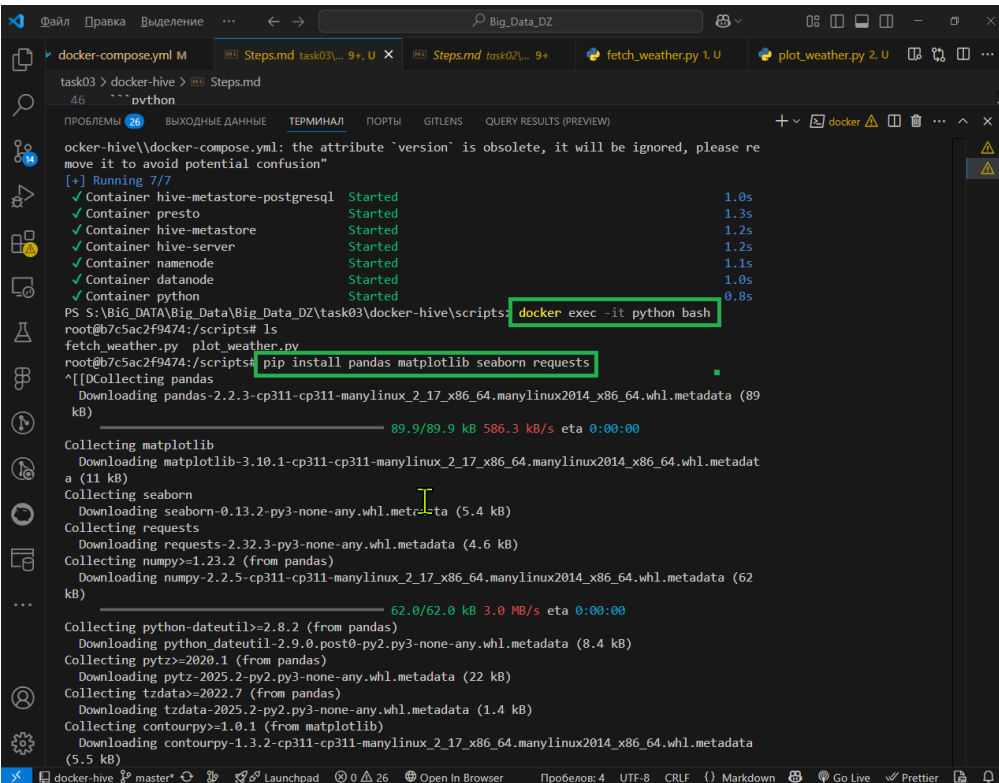
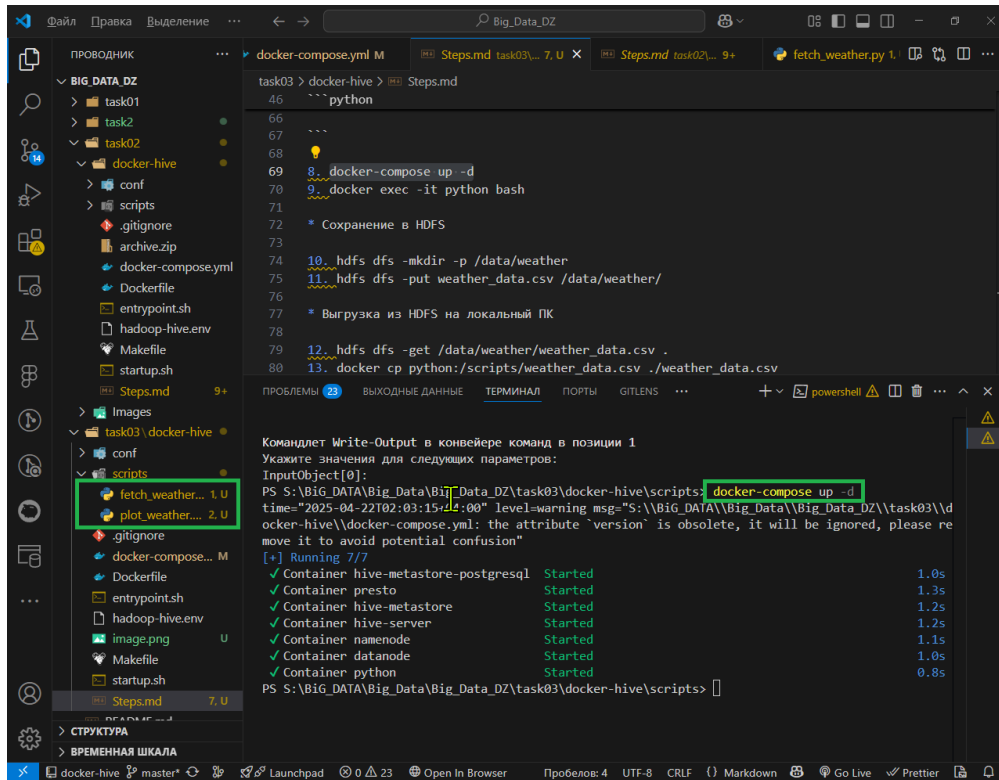
```
task03 > docker-hive > scripts > fetch_weather.py > ...
1 # scripts/fetch_weather.py
2 import requests
3 import pandas as pd
4 from datetime import datetime, timedelta
5
6 cities = {
7     "New York": (40.7128, -74.0060),
8     "London": (51.5074, -0.1278),
9     "Tokyo": (35.6895, 139.6917),
10    "Sydney": (-33.8688, 151.2093),
11    "Moscow": (55.7558, 37.6173),
12    "Baku": (40.409264, 49.867092)
13 }
14
15 end_date = datetime.today()
16 start_date = end_date - timedelta(days=30)
17
18 df_list = []
19 for city, (lat, lon) in cities.items():
20     url = f"https://archive-api.open-meteo.com/v1/archive?latitude={lat}&longitude={lon}&start_date={start_date}&end_date={end_date}"
21     res = requests.get(url).json()
22     for date, temp in zip(res["daily"]["time"], res["daily"]["temperature_2m_max"]):
23         df_list.append({
24             "city": city,
25             "date": date,
26             "temperature": temp
27         })
28
29 df = pd.DataFrame(df_list)
30 df.to_csv("weather_data.csv", index=False)
31 print("Weather data saved to weather_data.csv")
32
```

The terminal at the bottom shows an error message: "Не удается инициализировать устройство PRN" (Cannot initialize device PRN).

The screenshot shows the VS Code editor interface with the file explorer on the left. The main editor window shows the `plot_weather.py` script. The script reads the `weather_data.csv` file and creates two plots: a line plot showing temperature over time for each city, and a histogram showing the distribution of temperatures. The plots are saved as `weather_by_city.png` and `temperature_distribution.png`.

```
task03 > docker-hive > scripts > plot_weather.py > ...
1 # scripts/plot_weather.py
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 df = pd.read_csv("weather_data.csv")
7
8 plt.figure(figsize=(12, 6))
9 sns.lineplot(data=df, x="date", y="temperature", hue="city")
10 plt.title("Температура по городам за последний месяц")
11 plt.xticks(rotation=45)
12 plt.tight_layout()
13 plt.savefig("weather_by_city.png")
14 plt.clf()
15
16 sns.histplot(df["temperature"], bins=30, kde=True)
17 plt.title("Распределение температур")
18 plt.savefig("temperature_distribution.png")
19 print("Графики сохранены")
20
```

The terminal at the bottom shows the command `echo > plot_weather.py` being executed, which results in an error: "Командлет Write-Output в конвейере команд в позиции 1. Укажите значения для следующих параметров: InputObject[0]:".



File Editor: Big\_Data\_DZ

Steps.md 9+, U | fetch\_weather.py 1. U | plot\_weather.py 2. U

task03 > docker-hive > scripts > plot\_weather.py > ...

```

10 plt.title("Температура по городам за последний месяц")
11 plt.xticks(rotation=45)
12 plt.tight_layout()
13 plt.savefig("weather_by_city.png")
14 plt.clf()
15
16 sns.histplot(df["temperature"], bins=30, kde=True)
17 plt.title("Распределение температуры")
18 plt.savefig("temperature_distribution.png")

```

PROBLEMY 13 | ВЫХОДНЫЕ ДАННЫЕ | ТЕРМИНАЛ | ПОРТЫ | GITLENS

Running 7/7

- Container presto Running 0.0s
- Container python Running 0.0s
- Container hive-server Running 0.0s
- Container hive-metastore-postgresql Running 0.0s
- Container datanode Running 0.0s
- Container namenode Running 0.0s
- Container hive-metastore Running 0.0s

PS S:\Big\_Data\Big\_Data\Big\_Data\_DZ\task03\docker-hive\scripts> docker-compose up -d

time="2025-04-22T02:26:52+04:00" level=warning msg="S:\\Big\_Data\\Big\_Data\\Big\_Data\_DZ\\task03\\docker-hive\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"

Running 7/7

- Container hive-metastore-postgresql Started 0.5s
- Container presto Started 0.5s
- Container hive-server Started 0.4s
- Container namenode Started 0.7s
- Container python Started 0.6s
- Container hive-metastore Started 0.6s
- Container datanode Started 0.8s

PS S:\Big\_Data\Big\_Data\Big\_Data\_DZ\task03\docker-hive\scripts> docker exec -it python bash

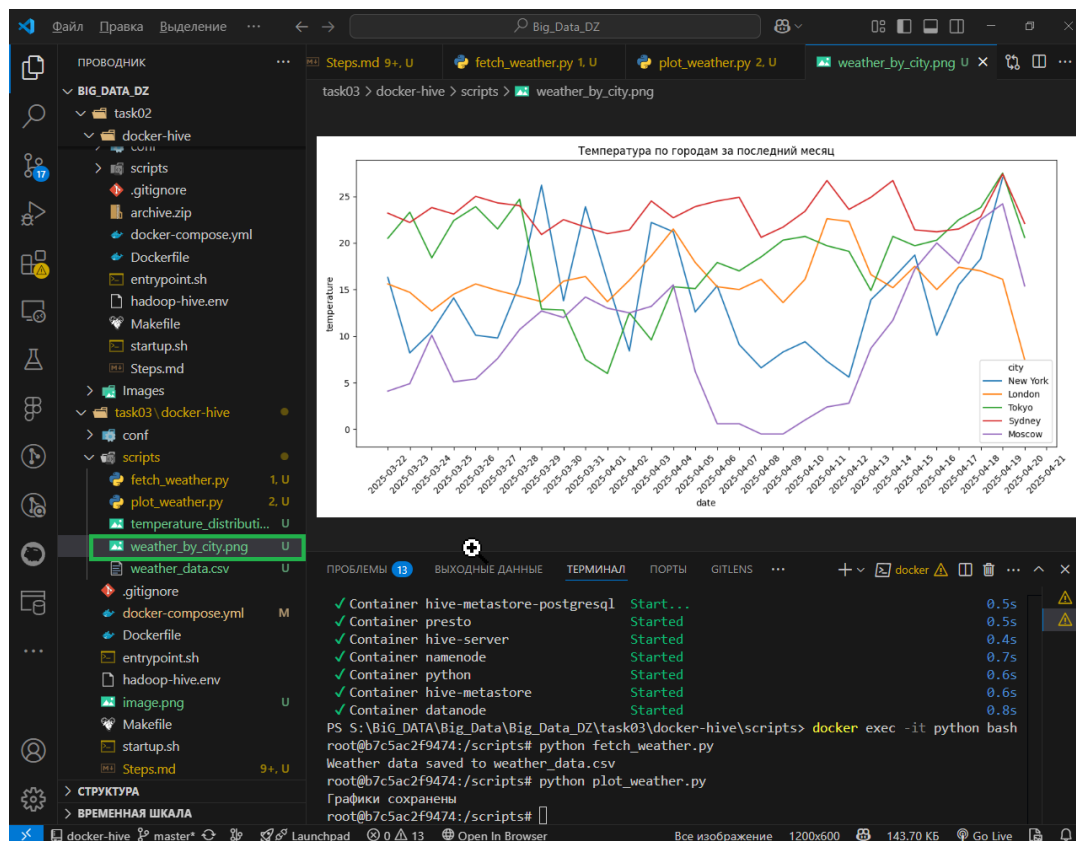
root@b7c5ac2f9474:/scripts# python fetch\_weather.py

Weather data saved to weather\_data.csv

root@b7c5ac2f9474:/scripts# python plot\_weather.py

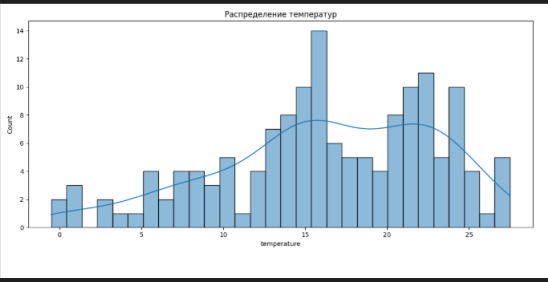
Графики сохранены

root@b7c5ac2f9474:/scripts#



Big\_Data\_DZ

task03 > docker-hive > scripts > temperature\_distribution.png



ПРОБЛЕМЫ 21 ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS ...

```
PS S:\Big_Data\Big_Data\Big_Data_DZ\task03\docker-hive> docker ps
```

| CONTAINER ID | IMAGE                                           | STATUS                  | PORTS                                        | COMMAND                   | NAMES            | CREATED  |
|--------------|-------------------------------------------------|-------------------------|----------------------------------------------|---------------------------|------------------|----------|
| b7c5ac2f9474 | python:3.11-slim                                | Up 11 minutes           |                                              | "tail -f /dev/null"       | python           | 34 минут |
| 098cf7dd1eea | bde2020/hive:2.3.2-postgresql-metastore         | Up 11 minutes           | 0.0.0.0:10000->10000/tcp, 10002/tcp          | "entrypoint.sh /bin/..."  | hive-server      | 2 days a |
| d24432e90c63 | bde2020/hive:2.3.2-postgresql-metastore         | Up 11 minutes           | 10000/tcp, 0.0.0.0:9083->9083/tcp, 10002/tcp | "entrypoint.sh /opt/..."  | hive-metastore   | 2 days a |
| f43207fb3209 | bde2020/hadoop-namenode:2.0.0-hadoop2.7.4-java8 | Up 11 minutes (healthy) | 0.0.0.0:50070->50070/tcp                     | "./bin/launcher run"      | namenode         | 2 days a |
| 4324ab4adb8b | shawnzhu/prestodb:0.181                         | Up 11 minutes           | 0.0.0.0:8080->8080/tcp                       | "./bin/launcher run"      | presto           | 2 days a |
| 730dc23fe7ae | bde2020/hadoop-datanode:2.0.0-hadoop2.7.4-java8 | Up 11 minutes (healthy) | 0.0.0.0:50075->50075/tcp                     | "./entrypoint.sh /run..." | datanode         | 2 days a |
| c5ca1bb13c0e | bde2020/hive-metastore-postgresql:2.3.0         | Up 11 minutes           | 5432/tcp                                     | "./docker-entrypoint..."  | hive-metastore-p | 2 days a |

```
PS S:\Big_Data\Big_Data\Big_Data_DZ\task03\docker-hive> docker exec -it namenode bash
root@f43207fb3209:/#
```

Big\_Data\_DZ

task03 > docker-hive > Steps.md

```
72 11. python fetch_weather.py
73 12. python plot_weather.py
74 13. exit
75 14. docker exec -it namenode bash
76 15. hdfs dfs -ls /
77 16. hdfs dfs -mkdir -p /data/weather
78 17. hdfs dfs -put weather_data.csv /data/weather/
79
```

ПРОБЛЕМЫ 21 ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS ...

```
PS S:\Big_Data\Big_Data\Big_Data_DZ\task03\docker-hive> docker exec -it namenode bash
root@f43207fb3209:/# hdfs dfs -ls /
Found 4 items
drwxr-xr-x - root supergroup 0 2025-04-18 23:53 /data
drwxr-xr-x - root supergroup 0 2025-04-19 00:17 /output
drwxrwxr-x - root supergroup 0 2025-04-18 23:42 /tmp
drwxr-xr-x - root supergroup 0 2025-04-18 23:42 /user
root@f43207fb3209:/#
```

```
task03 > docker-hive > Steps.md
69 8. docker-compose up -d
70 9. docker exec -it python bash
71 10. pip install pandas matplotlib seaborn requests
72 11. python fetch_weather.py
73 12. python plot_weather.py
74 13. exit
75 14. docker cp scripts/weather_data.csv namenode:/tmp/
76 15. docker exec -it namenode bash
77 16. hdfs dfs -ls /
78 17. hdfs dfs -mkdir -p /data/weather
79 18. hdfs dfs -put /tmp/weather_data.csv /data/weather/
80

ПРОБЛЕМЫ (30) ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS QUERY RESULTS (PREVIEW) + powershell

exit
PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker exec -it python bash
PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker exec -it python bash
root@b7c5ac2f9474:/scripts# ls
fetch_weather.py temperature_distribution.png weather_data.csv
fetch_weather.py temperature_distribution.png weather_data.csv
plot_weather.py weather_by_city.png
root@b7c5ac2f9474:/scripts# exit
exit
exit
PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker cp scripts/weather_data.csv namenode:/data/weather/
no such directory
PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker cp scripts/weather_data.csv namenode:/tmp
Successfully copied 5.12kB to namenode:/tmp
PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker cp scripts/weather_data.csv namenode:/tmp
Successfully copied 5.12kB to namenode:/tmp
PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive>
```

```
task03 > docker-hive > Steps.md
69 8. docker-compose up -d
70 9. docker exec -it python bash
71 10. pip install pandas matplotlib seaborn requests
72 11. python fetch_weather.py
73 12. python plot_weather.py
74 13. exit
75 14. docker cp scripts/weather_data.csv namenode:/tmp/
76 15. docker exec -it namenode bash
77 16. hdfs dfs -ls /
78 17. hdfs dfs -mkdir -p /data/weather
79 18. hdfs dfs -put /tmp/weather_data.csv /data/weather/
80

ПРОБЛЕМЫ (30) ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS QUERY RESULTS (PREVIEW) + docker

PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker cp scripts/weather_data.csv namenode:/tmp
Successfully copied 5.12kB to namenode:/tmp
PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker cp scripts/weather_data.csv namenode:/tmp
DZ\task03\docker-hive>
Successfully copied 5.12kB to namenode:/tmp/
PS S:\BIG_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker exec -it namenode bash
root@f43207fb3209:/# hdfs dfs -put /tmp/weather_data.csv /data/weather/
root@f43207fb3209:/# hdfs dfs -ls /
Found 4 items
drwxr-xr-x - root supergroup 0 2025-04-21 22:45 /data
drwxr-xr-x - root supergroup 0 2025-04-19 00:17 /output
drwxrwxr-x - root supergroup 0 2025-04-18 23:42 /tmp
drwxr-xr-x - root supergroup 0 2025-04-18 23:42 /user
root@f43207fb3209:/# hdfs dfs -ls /data/
Found 2 items
drwxr-xr-x - root supergroup 0 2025-04-19 21:54 /data/airbnb
drwxr-xr-x - root supergroup 0 2025-04-21 23:01 /data/weather
root@f43207fb3209:/# hdfs dfs -ls /data/weather/
Found 1 items
-rw-r--r-- 3 root supergroup 3569 2025-04-21 23:01 /data/weather/weather_data.csv
root@f43207fb3209:/#
```



```
SUGARFREE :: Togliersi di torno) :: 0:00
task03 > docker-hive > Steps.md
88 * Выгружаем обратно на локальную машину
89
90 20. docker exec -it namenode bash
91 21. mkdir -p /scripts
92 22. hdfs dfs -get /data/weather/weather_data.csv /scripts/weather_data_from_hdfs.csv
93 23. docker cp namenode:/scripts/weather_data_from_hdfs.csv ./scripts/weather_data_from_hdfs.csv
94
95 20. hdfs dfs -get /data/weather/weather_data.csv /scripts/weather_data_from_hdfs.csv

ПРОБЛЕМЫ 38 Выходные данные ТЕРМИНАЛ ПОРТЫ GITLINS QUERY RESULTS (PREVIEW) + powershell

root@f43207fb3209: # mkdir -p /scripts
root@f43207fb3209: # ls
bin dev etc hadoop-data lib media opt root run.sh scripts sys usr
boot entrypoint.sh hadoop_home lib64 mnt proc run sbin srv tmp var
root@f43207fb3209:/# hdfs dfs -ls /
Found 4 items
drwxr-xr-x - root supergroup 0 2025-04-21 22:45 /data
drwxr-xr-x - root supergroup 0 2025-04-19 00:17 /output
drwxr-xr-x - root supergroup 0 2025-04-18 23:42 /tmp
drwxr-xr-x - root supergroup 0 2025-04-18 23:42 /user
root@f43207fb3209:/# hdfs dfs -get /data/weather/weather_data.csv /scripts/weather_data_from_hdfs.csv
root@f43207fb3209:/# docker cp namenode:/scripts/weather_data_from_hdfs.csv ./scripts/weather_data_from_hdfs.csv
bash: docker: command not found
root@f43207fb3209:/# exit
exit
PS S:\Big_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker cp namenode:/scripts/weather_data_from_hdfs.csv ./scripts/weather_data_from_hdfs.csv
Successfully copied 5.12kB to S:\Big_DATA\Big_Data\Big_Data_DZ\task03\docker-hive\scripts\weather_data_from_hdfs.csv
PS S:\Big_DATA\Big_Data\Big_Data_DZ\task03\docker-hive>
```

```
PROБЛЕМЫ 21 Выходные данные ТЕРМИНАЛ ПОРТЫ ... powershell
task03 > docker-hive > Steps.md
73 12. python plot_weather.py
74 13. exit
75
76 * Сохранить результаты в HDFS
77
78 14. docker cp scripts/weather_data.csv namenode:/tmp/
79 15. docker exec -it namenode bash
80 16. hdfs dfs -ls /
81 17. hdfs dfs -mkdir -p /data/weather
82 18. hdfs dfs -put /tmp/weather_data.csv /data/weather/
83
84 * Проверка – убедись, что файл в HDFS
85
86 19. hdfs dfs -ls /data/weather
87
88 * Выгружаем из HDFS обратно на локальную машину
89
90 20. docker exec -it namenode bash
91 21. mkdir -p /scripts
92 22. hdfs dfs -get /data/weather/weather_data.csv /scripts/
93 23. exit
94 24. docker cp namenode:/scripts/weather_data_from_hdfs.csv ./scripts/
weather_data_from_hdfs.csv

root@f43207fb3209:/# docker cp namenode:/scripts/weather_data_from_hdfs.csv ./scripts/
root@f43207fb3209:/# docker cp namenode:/scripts/weather_data_from_hdfs.csv ./scripts/
bash: docker: command not found
root@f43207fb3209:/# exit
exit
PS S:\Big_DATA\Big_Data\Big_Data_DZ\task03\docker-hive> docker cp namenode:/scripts/weather_data_from_hdfs.csv ./scripts/
weather_data_from_hdfs.csv
Successfully copied 5.12kB to S:\Big_DATA\Big_Data\Big_Data_DZ\task03\docker-hive\scripts\weather_data_from_hdfs.csv
PS S:\Big_DATA\Big_Data\Big_Data_DZ\task03\docker-hive>
```